

# **Low Contrast Detectability for CT Toolbox**

## User Manual

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## Tool Summary

**Low Contrast Detectability for CT (LCD-CT) Toolbox** provides a common interface to evaluate the low contrast detectability (LCD) performance of advanced nonlinear CT image reconstruction and denoising algorithms. The toolbox uses model observers (MO) to evaluate the LCD of targets with known locations in test images obtained with the standard uniform-background *MITA-LCD phantom* or a nonuniform-background Liver-LCD phantom. The model observer detection accuracy is measured by the area under the receiver operating characteristic curve (AUC) and the detectability signal-to-noise ratio ( $d_{snr}$ ). The LCD-CT toolbox can be used by CT developers to perform initial evaluation on image quality improvement or dose reduction potential of their reconstruction and denoising algorithms.

The code is affiliated with the CDRH Regulatory Science Tools *LCD-CT* and LCD-CT-v2 ([placeholder link](#)).

## Disclaimer

### About the Catalog of Regulatory Science Tools

The enclosed tool is part of the *Catalog of Regulatory Science Tools*, which provides a peer-reviewed resource for stakeholders to use where standards and qualified Medical Device Development Tools (MDDTs) do not yet exist. These tools do not replace FDA-recognized standards or MDDTs. This catalog collates a variety of regulatory science tools that the FDA's Center for Devices and Radiological Health's (CDRH) Office of Science and Engineering Labs (OSEL) developed. These tools use the most innovative science to support medical device development and patient access to safe and effective medical devices. If you are considering using a tool from this catalog in your marketing submissions, note that these tools have not been qualified as *Medical Device Development Tools* and the FDA has not evaluated the suitability of these tools within any specific context of use. You may [request feedback or meetings for medical device submissions](#) as part of the Q-Submission Program.

For more information about the Catalog of Regulatory Science Tools, email [RST\\_CDRH@fda.hhs.gov](mailto:RST_CDRH@fda.hhs.gov).

## Main Features

The LCD-CT toolbox has three main features corresponding to the LCD-CT test workflow (Fig. 1).

1. Digital phantom and CT simulation:

- Creating digital replica of the background and signal modules of the [MITA-LCD phantom](#) and a digital Liver-LCD phantom that contains low-contrast disks in a non-uniform, anatomical background.
- Simulating sinogram and generate fan-beam CT scans of the digital phantoms based on the publicly available [Michigan Image Reconstruction Toolbox \(MIRT\)](#). Note that the MIRT package is compatible with Linux and mac systems but may not be fully compatible with Windows (see [MIRT Readme](#) ).

*(This feature runs best in MATLAB.)*

## 2. LCD test:

- Estimating low contrast detectability performance from the MITA-LCD or Liver-LCD phantom CT images using channelized Hotelling model observer with Laguerre-Gauss (LG) channels and two options of Difference-of-Gaussian (DOG) channels and Gabor channels.

*(This feature runs in both MATLAB and python.)*

## 3. Dose reduction estimation:

- Estimating the dose reduction percentages of an evaluated nonlinear reconstruction method relative to a reference method (e.g., filtered back projection method) using the AUC results of the evaluated and the reference reconstruction methods across multiple dose levels obtained from a LCD test.

*(This feature runs only in python.)*

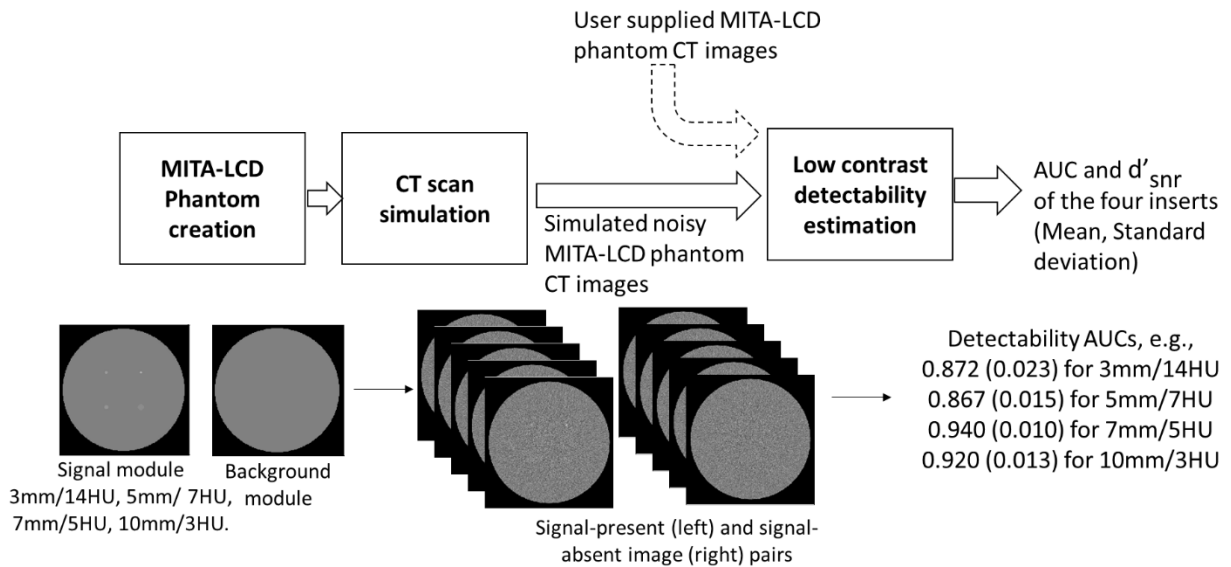


Figure 1: Illustration of the LCD-CT workflow, including phantom creation, CT simulation and low-contrast detectability estimation to produce an output of AUC measurements associated with each low-contrast disk.

## Start Here

### Requirements

- **Python (>= 3.8)** with packages listed in pyproject.toml (numpy, scipy, scikit-image, etc.)
- **Matlab (version > R2016a)**

### Installation

1. Git clone the LCD-CT Toolbox repository:

```
git clone https://github.com/DIDSR/LCD_CT
cd LCD_CT
```

2. **Python:**

- Create a [conda](#) environment and install the package:

```
conda env create --file environment.yml conda activate LCD_CT pip install -e .
```

```
conda create -n LCD_CT python=3.8 pip -y
```

```
conda activate LCD_CT
```

```
conda install -c conda-forge -c defaults octave cxx-compiler pandas tomli numpy oct2py pytest simpleitk scikit-image scikit-learn scipy matplotlib sphinx-tabs pandoc ipykernel git -y
```

```
pip install "git+https://github.com/DIDSR/pediatricIQphantoms" sphinxcontrib-svg2pdfconverter nbsphinx
```

```
pip install -e .
```

*Expected run time: 2-5 min*

- Test the python installation

Run the following tests in the conda LCD-CT virtual environment:

```
pytest tests/test_lcd.py
```

```
python demo_analyze_dose_reduction.py
```

3. **MATLAB:**

MATLAB version information is available from [MATLAB](#). See how to get [matlab version](#) on your system.

From the bash command line

```
matlab -batch "test"
```

Or, from the Matlab prompt

```
>> test
```

*Expected run time: 2 min 30 s*

4. **(Optional) GNU OCTAVE:**

If Matlab is not available, [GNU Octave](#) (**version > 4.4**) can be installed using `source install.sh` to prepare a [conda](#) environment.

Please note that the MATLAB part of this software has been fully developed and tested using MATLAB. GNU Octave may also be used to run the software; however, Octave compatibility has not been fully tested or validated. Although Octave is largely compatible with MATLAB, differences exist in the availability and behavior of certain functions and toolboxes. Users who choose to run this software with GNU Octave may need to install additional Octave packages and modify or replace MATLAB-specific functions with their Octave-compatible equivalents. Users are responsible for making any necessary adaptations for their specific Octave environment and for verifying that the resulting outputs are consistent with their expected use.

- Create a Conda Octave environment and install the package:

```
source install.sh
```

*Expected run time: 10-30 min*

- Test Octave

In the Conda Octave virtual environment, run:

```
octave test.m
```

\*Expected run time: 2 min.

**Note:** The Liver-LCD Phantom Creation code (`makeCT_LiverLCD.m` in folder "LCD Phantom creation") is not compatible with Octave, because it calls mex-functions in MIRT that are not compiled for Octave (check [MIRT webpage](#)). For this reason, `makeCT_LiverLCD` is excluded from being tested in Octave. Use MATLAB to run `makeCT_LiverLCD.m`.

## How to Use the LCD-CT Toolkit

After installing, explore the demos to learn how to use the tool to assess low contrast detectability:

### Python Demos

- LCD test demos: These demos are intended to be run linearly and demonstrate the use of the LCD-CT tool and how it can be used in more sophisticated loops to understand LCD relationships with different imaging conditions, lesions, and model observer types.
  - [demo\\_01\\_singlerecon\\_LCD.py](#)
  - [demo\\_02\\_tworecon\\_LCD.py](#)
  - [demo\\_03\\_tworecon\\_dosecurve\\_LCD.py](#)

- Dose reduction potential estimation demo: this demo shows how to estimate dose reduction percentages of an evaluated reconstruction method relative to a reference FBP method using the [dose, AUC] values obtained in the LCD test across multiple dose levels.
  - [demo\\_analyze\\_dose\\_reduction.py](#)

## MATLAB Demos (Octave as optional)

- LCD test demos: These demos are intended to be run linearly and demonstrate the use of the LCD-CT tool and how it can be used in more sophisticated loops to understand LCD relationships with different imaging conditions, lesions, and model observer types.
 

(demo using MITA-LCD images)

  - [demo\\_01\\_singlerecon\\_LCD.m](#)
  - [demo\\_02\\_tworecon\\_LCD.m](#)
  - [demo\\_03\\_tworecon\\_dosecurve\\_LCD.m](#)

(demo using Liver-LCD images)

  - [demo\\_liver\\_lcd\\_test.m](#)
- Phantom creation demo: These demo run makeCT\_CCT189 and makeCT\_LiverLCD to shows how to simulated CT noisy CT images of the digital MITA-LCD and Liver-LCD phantoms using the Michigan Image Reconstruction Toolkit .
  - [demo\\_test\\_phantomcreation.m](#) Expected run time (Matlab): 60 s

### Note on the use of Michigan Image Reconstruction Toolkit

The LCD Phantom Creation code uses functions from *Michigan Image Reconstruction Toolkit (MIRT)* (<https://web.eecs.umich.edu/~fessler/irt/fessler.tgz>). It should be automatically downloaded and installed when 'demo\_test\_phantomcreation.m' is run. If the automatic download does not work (this can happen when the matlab/octave unzip() function does not successfully extract all the files), this can be done manually:

- 1) download MIRT from <https://github.com/JeffFessler/mirt>;
- 2) Unzip MIRT to a local directory.
- 3) In Matlab, Run the file "setup.m" in the MIRT local directory to add all the MIRT subdirectories to the MATLAB workspace.

To test whether the manual setup is successful, run **demo\_test\_phantomcreation.m**.

### Note on accessing the Large Dataset

By default the supplied LCD test demos and test scripts will use the [small dataset](#) included in the repository. The boolean `use_large_dataset` is set to False by default [https://github.com/DIDSR/LCD\\_CT/blob/aa9a907376a413d2b87c434590c033d812e2ac75/test.m#L6](https://github.com/DIDSR/LCD_CT/blob/aa9a907376a413d2b87c434590c033d812e2ac75/test.m#L6). Changing the variable to true followed by rerunning any of the scripts will download from [Zenodo](#) the large dataset (1.3 GB)(if not automatically, user can manually download the Zenodo dataset. Data link provided in the demos) and use it in subsequent analyses.

Additional demos of tool usage can be found in [additional\\_demos](#).

## Tool Reference

- RST Reference Number: RST24MD08.01
- Date of Publication: 09/24/2023
- Recommended Citation:
  - U.S. Food and Drug Administration. (2023). LCD-CT: Low-contrast Detectability (LCD) Test for Assessing Advanced Nonlinear CT Image Reconstruction and Denoising Methods (RST24MD08.01). <https://cdrh-rst.fda.gov/lcd-ct-low-contrast-detectability-lcd-test-assessing-advanced-nonlinear-ct-image-reconstruction-and>
  - U.S. Food and Drug Administration. (202x). LCD-CT-v2: Low-contrast Detectability (LCD) Test for Assessing Advanced Nonlinear CT Image Reconstruction and Denoising Methods (v2) (*RSTxxxx.xx to be determined*).
  - Rongping Zeng, Prabhat Kc, “Factors Influencing LCD Test-based Dose Reduction Potential Estimates for Deep Learning Image Reconstruction and Denoising Methods in CT”, ([link to be added](#))